US Equity Momentum and Inverse-Volatility Portfolios

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Abstract

This study rebuilds, from monthly prices of 100 US large-cap stocks between December 2002 and August 2018, a decile momentum strategy and an inverse-volatility portfolio measured against an equal-weight benchmark. Over this sample the momentum long-short lost -0.45% per month and is not statistically distinguishable from zero. It carries a large loading on the published momentum factor yet a significant negative four-factor alpha, so a ten-percent breakpoint equal-weighted construction is a weaker version of the factor. The inverse-volatility portfolio tracked the equal-weight benchmark closely, with an active return that was small and insignificant. The results survive alternative formation windows, quintile sorting, and transaction costs.

1 Introduction

Momentum, the tendency of recent winners to keep outperforming recent losers, is one of the most studied anomalies in asset pricing (Jegadeesh and Titman, 1993; Carhart, 1997). Risk-based weighting, which tilts a portfolio toward lower-volatility names, is a common alternative to market-capitalisation or equal weighting. This study examines both on a single panel of 100 US large-cap stocks.

Three questions are addressed. First, a momentum strategy is built by sorting stocks into deciles on their past return and holding the top decile against the bottom decile. Second, an inverse-volatility portfolio is compared with an equal-weight benchmark, and the active return between them is decomposed with a rolling Fama-French regression. Third, a short identity for the volatility contribution of a factor is derived. The headline finding is that momentum did not pay in this sample: average returns are flat to slightly declining across the deciles, and the long-short was driven deeply negative by the rebound of April 2009, consistent with the momentum crashes documented by Daniel and Moskowitz (2016).

2 Data

The universe is 100 US large-cap stocks. Monthly closing prices in US dollars, including dividends, run from November 2002 to August 2018, which gives 189 monthly returns from December 2002. Returns are arithmetic, $r_{i,t} = P_{i,t}/P_{i,t-1} - 1$.

Factor returns are the Kenneth French North American market, size, and value factors, the risk-free rate, and the North American momentum factor (Fama and French, 1993, 2012; French). The series pasted into the original workbook are used for the replication, so the rebuilt numbers can be checked against the original cell by cell. A script also downloads the current factor files and records the download date in the repository. The current files differ from the 2018 copies by up to 0.73 percentage points on value, 0.28 on size, and 0.18 on the market in individual months, which reflects the routine revisions Kenneth French applies to the North American history; the risk-free rate is unchanged.

3 Method

3.1 Momentum

For each month the formation return is the twelve-month price change that skips the most recent month, $F_{i,t} = P_{i,t-2}/P_{i,t-14} - 1$. The skip avoids the short-term reversal in the latest month. Stocks are ranked on $F_{i,t}$ and split into ten deciles of ten stocks. Each decile return is the equal-weight average of its stocks' returns in the holding month, and the strategy return is the top decile minus the bottom decile.

3.2 Inverse-volatility and equal-weight portfolios

Each stock's volatility is the sample standard deviation of its returns over the trailing twelve months, excluding the current month. The inverse-volatility weight is $w_{i,t} = (1/\sigma_{i,t})/\sum_j (1/\sigma_{j,t})$, and the portfolio return is $\sum_i w_{i,t} r_{i,t}$. The equal-weight benchmark holds one percent in each stock. The active return is the inverse-volatility return minus the equal-weight return.

3.3 Rolling factor model

The active return is regressed on the market, size, and value factors over a rolling twelve-month window by ordinary least squares with an intercept. The estimated betas are applied to the current month's factors to split the active return into a systematic part, $\alpha + \sum_k \beta_k f_{k,t}$, and an idiosyncratic residual. Statistical significance uses Newey-West standard errors (Newey and West, 1987).

3.4 A volatility contribution identity

Writing a portfolio return as a sum of components $r_i = a_i + \beta_{i,p} f_i + \varepsilon_i$ and using $\operatorname{cov}(X, Y) = \sigma(X)\,\sigma(Y)\,\operatorname{corr}(X, Y)$,

$$\sigma^2(r_p) = \operatorname{cov}\Big(r_p, \sum_{i=1}^{N} r_i\Big) = \sigma(r_p)\,\sigma\Big(\sum_i r_i\Big)\operatorname{corr}\Big(r_p, \sum_i r_i\Big). \tag{1}$$

Dividing by $\sigma(r_p)$ and expanding each component, the constant terms have zero volatility and zero correlation with the portfolio, which leaves the contribution of each factor to portfolio volatility,

$$\sigma(r_p) = \sum_{i=1}^{N} \beta_{i,p}\,\sigma(f_i)\,\operatorname{corr}(r_p, f_i). \tag{2}$$

4 Results

4.1 Momentum

The momentum long-short returned -0.45% per month over 176 months, an annualised -5.42% with volatility 30.42%, so the Sharpe ratio is -0.18. The Newey-West t-statistic is -0.65, so the strategy is not distinguishable from zero. Average returns do not rise across the deciles: the loser decile

averaged 1.08% per month against 0.63% for the winner decile (Figure 1). The worst month was -56.88% in April 2009, when prior losers rebounded sharply, and the strategy's cumulative return over the sample was -80.5% (Figure 3). The strategy moves closely with the published momentum factor, with a correlation of 0.83 (Figure 2).

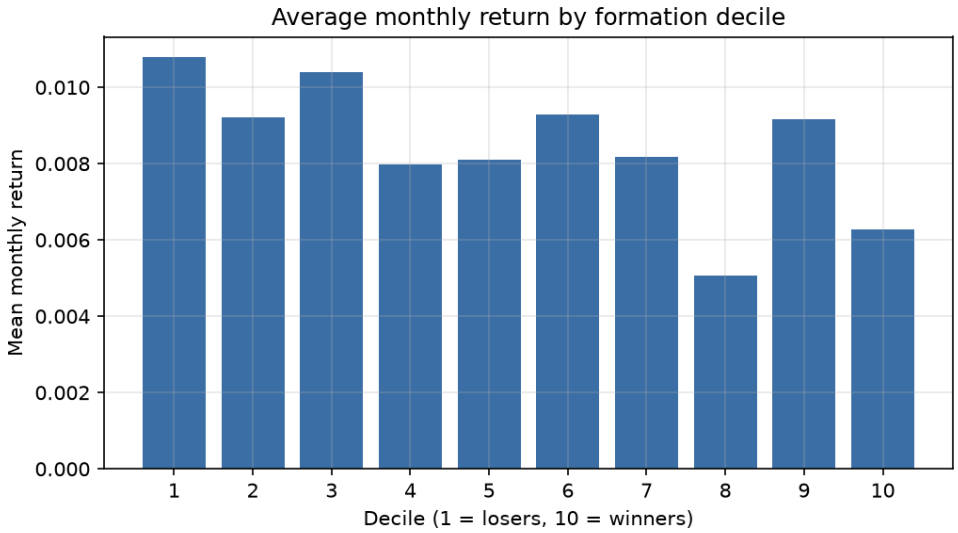


Figure 1: Average monthly return by formation decile. Decile 1 holds the lowest past returns and decile 10 the highest.

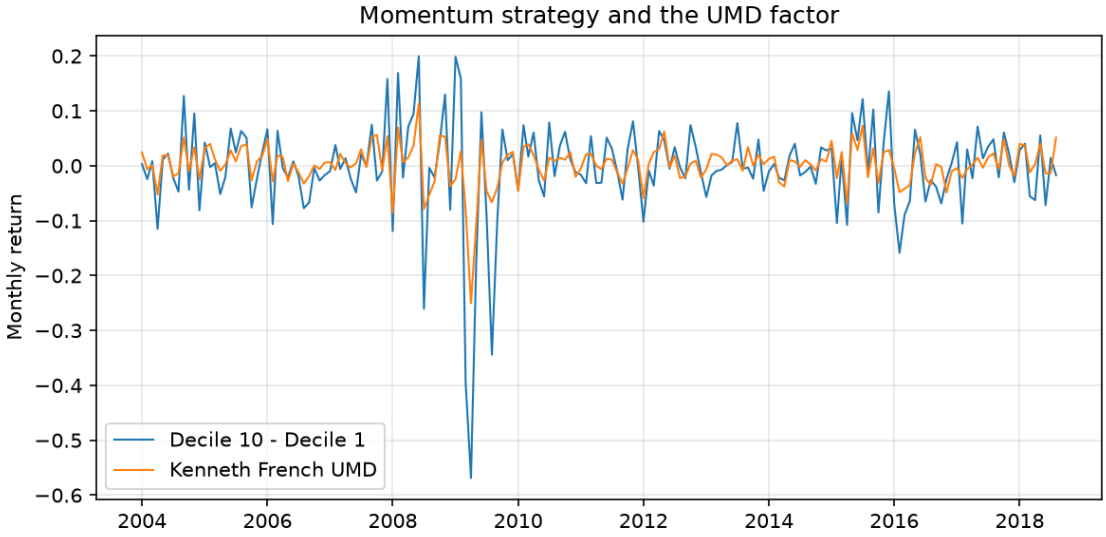


Figure 2: The decile momentum long-short and the Kenneth French momentum factor.

4.2 Inverse-volatility and equal-weight portfolios

The inverse-volatility portfolio returned 0.80% per month with a Sharpe ratio of 0.78, close to the equal-weight benchmark at 0.87% and 0.74 (Table 1, Figure 4). The active return was -0.07% per

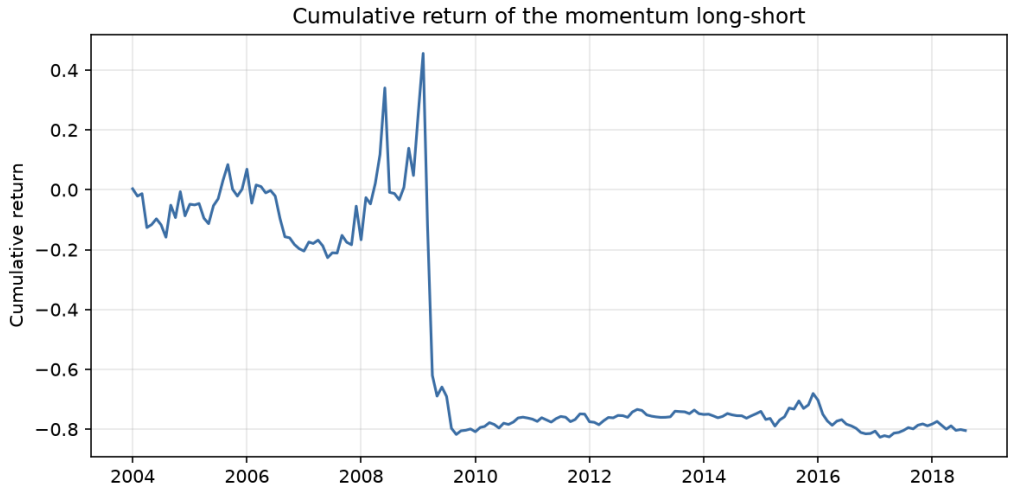


Figure 3: Cumulative return of the momentum long-short, including the April 2009 crash.

month with a Newey-West t-statistic of -1.04, so the two portfolios are hard to tell apart. The rolling betas of the active return move over time and average -0.06 on the market, -0.08 on size, and -0.05 on value (Figure 5). On average 0.61 of the variance of the active return is explained by the three factors, so a large part remains idiosyncratic (Figures 6 and 7).

Table 1: Portfolio statistics. Mean is per month, volatility is annualised, and the t-statistic uses Newey-West standard errors.

Portfolio	Mean (mo)	Vol (yr)	Sharpe	NW t	Cumulative
Inverse volatility	0.80%	12.21%	0.78	2.73	265%
Equal weight	0.87%	14.13%	0.74	2.53	301%
Active (IV - EW)	-0.07%	2.97%	-0.30	-1.04	-13%

5 Robustness

Formation window. Three formation windows give the same message (Table 2). Dropping the skip, or shortening the window to six months, makes the long-short more negative, not less, so the skip-month twelve-month definition is the least weak of the three and none of them pays.

Table 2: Momentum long-short under alternative formation windows.

Formation window	Months	Mean (per month)	NW t	Sharpe
12 months, skip most recent	176	-0.45%	-0.65	-0.18
12 months, no skip	177	-0.55%	-0.81	-0.22
6 months, skip most recent	182	-0.68%	-1.10	-0.30

Quintiles. With only 100 stocks, quintiles of twenty names are less noisy than deciles. The quintile long-short returned -0.23% per month with a Newey-West t-statistic of -0.46, still indistin-

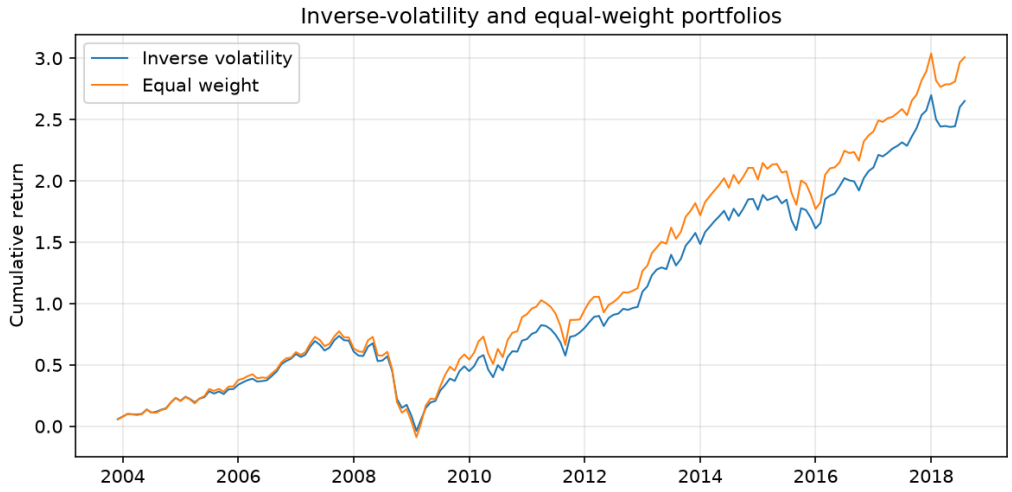


Figure 4: Cumulative return of the inverse-volatility and equal-weight portfolios.

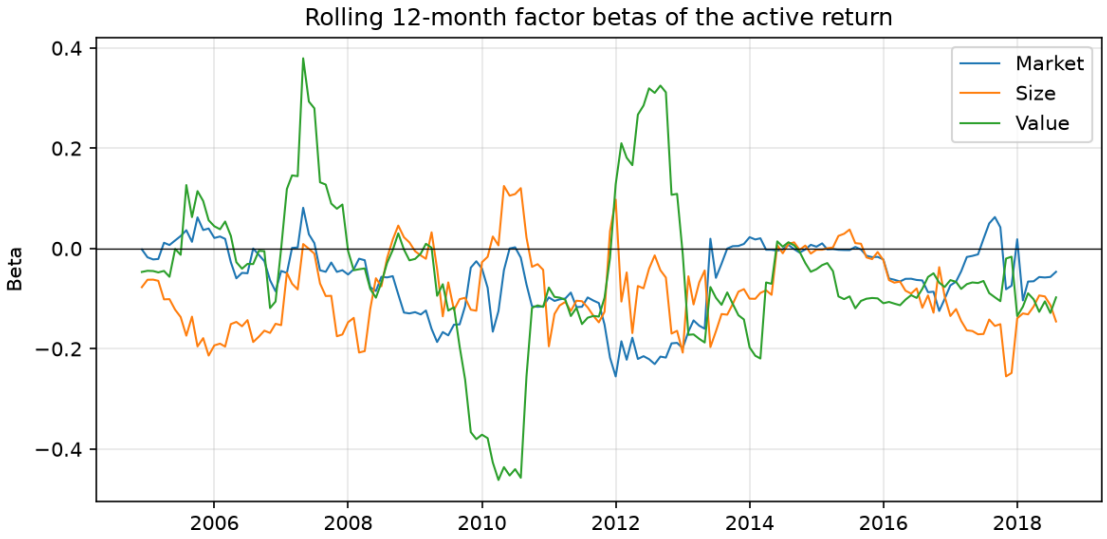


Figure 5: Rolling twelve-month factor betas of the active return.

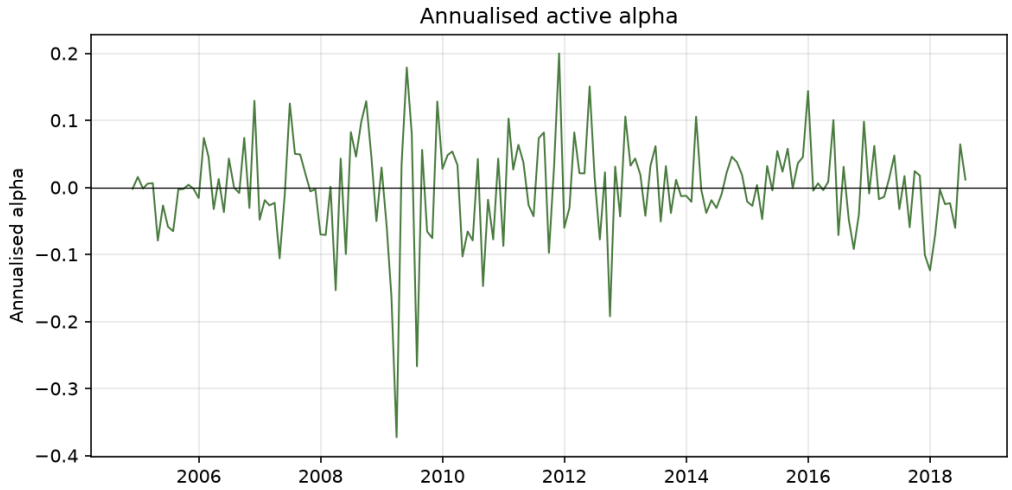


Figure 6: Annualised active alpha from the rolling regression.

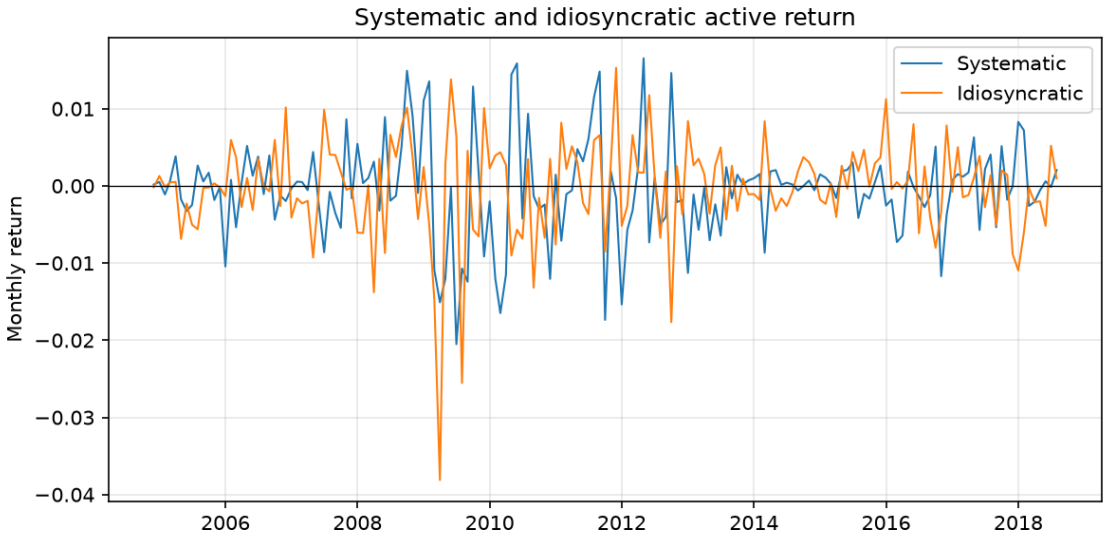


Figure 7: Systematic and idiosyncratic parts of the active return.

guishable from zero.

Factor alphas. Table 3 regresses the decile long-short on the CAPM, the Fama-French three-factor model, and the Carhart four-factor model. The market and three-factor alphas are small and insignificant. Under the four-factor model the strategy loads on the momentum factor with a beta of 1.84 (t 14.45), which lifts the fit to an R^2 of 0.70, yet the alpha turns significantly negative at -0.72% per month (t -2.22). The decile long-short is a weaker version of the published momentum factor, which uses wider breakpoints and value weights.

Table 3: Factor regressions of the momentum long-short. Alpha is per month; t-statistics use Newey-West standard errors.

Model	α (mo)	$t(\alpha)$	β_{mkt}	β_{smb}	β_{hml}	β_{umd}	R^2
CAPM	-0.02%	-0.04	-0.60	–	–	–	0.07
Fama-French 3	-0.18%	-0.33	-0.55	0.28	-1.32	–	0.19
Carhart 4	-0.72%	-2.22	-0.26	0.16	-0.19	1.84	0.70

Turnover and costs. The momentum long-short rebalances almost entirely each month, with two-way turnover of 115% per month, so costs matter: at fifty basis points per unit traded the mean return falls to -1.03% per month (Table 4). The inverse-volatility and equal-weight books trade far less, 7.9% and 4.5% per month, so costs barely move the active return. The equal-weight benchmark still turns over because it is rebalanced back to equal weights each month.

Table 4: Two-way turnover and mean net return after costs, per month.

Book	Turnover (mo)	Net at 0 bps	Net at 10 bps	Net at 50 bps
Momentum long-short	115%	-0.45%	-0.57%	-1.03%
Active (IV - EW)	7.9% / 4.5%	-0.07%	-0.08%	-0.09%

Replication. The Python rebuild matches the original workbook to floating-point precision across returns, deciles, portfolios, and the rolling regression. One isolated error was found: in April 2009's neighbour month a formation-return tie placed one stock twice in a middle decile and dropped another, which affected only that decile in that month and left the long-short unchanged. The comparison is documented in the repository.

6 Conclusion

On this panel of 100 US large caps, momentum did not pay between December 2002 and August 2018. The long-short was flat to negative, fragile to the April 2009 rebound, and delivered a significant negative alpha once the momentum factor is included, which marks it as a weaker version of the published factor rather than an independent source of return. Inverse-volatility weighting produced almost the same result as equal weighting, with a small and insignificant active return that the three factors explain only in part. The findings are stable across formation windows, quintile sorting, and reasonable transaction costs.

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